Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Further Trigonometry

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $3\sin 2\theta - 4\cos\theta = 0$ for $0 \le \theta \le 2\pi$. **[5]**

2. Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0 \le x \le 2\pi$. **[5]**

3. Solve $2\sin^2 x - \sin x - 1 = 0$ for $0 \le x \le 2\pi$. **[5]**

4. Solve $3\cos 2\theta + 2\cos\theta = 0$ for $0 \le \theta \le \pi$. **[5]**

5. Solve $\sin 2x = \cos x$ for $0 \le x \le 2\pi$. **[5]**

6. Prove $\dfrac{\sin 3\theta}{\sin\theta} = 3 - 4\sin^2\theta$ $(\sin\theta \neq 0)$. **[5]**

7. Prove $\cos 2\theta = 2\cos^2\theta - 1$. **[4]**

8. Prove $\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$. **[5]**

9. Solve $\sin 3\theta = 2\sin\theta$ for $0 \le \theta \le \pi$. **[5]**

10. Solve $\cos 2\theta = \sin\theta$ for $0 \le \theta \le 2\pi$. **[5]**

11. Solve $2\cos 2\theta - 3\cos\theta + 1 = 0$ for $0 \le \theta \le 2\pi$. **[6]**

12. Solve $4\sin^2\theta - 3\sin\theta = 0$ for $0 \le \theta \le 2\pi$. **[4]**

13. Solve $\tan 2\theta = \sqrt{3}$ for $0 \le \theta \le \pi$. **[4]**

14. Solve $\sin\theta + \cos\theta = 1$ for $0 \le \theta \le 2\pi$. **[5]**

15. Solve $2\sin\theta\cos\theta = \frac{1}{2}$ for $0 \le \theta \le 2\pi$. **[4]**

Section B – Hard (Q16–25)

16. Express $5\cos\theta - 12\sin\theta$ in the form $R\cos(\theta + \alpha)$. Hence solve $= 4$ for $0 \le \theta < 2\pi$. **[8]**

17. Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$. Hence solve $= 2$ for $0 \le \theta < 2\pi$. **[8]**

18. Express $8\cos\theta + 15\sin\theta$ in the form $R\cos(\theta - \alpha)$. Hence solve $= 10$. **[8]**

19. Solve $2\cos 2\theta + \cos\theta = 1$ for $0 \le \theta \le 2\pi$. **[6]**

20. Solve $\sin 2\theta + \cos\theta = 0$ for $0 \leq \theta \leq 2\pi$. **[6]**

21. Prove $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. **[5]**

22. Prove $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. **[5]**

23. Solve $\cos 3\theta = \cos\theta$ for $0 \leq \theta \leq \pi$. **[6]**

24. Find the general solution of $\sin 2\theta = \frac{1}{2}$. **[4]**

25. Find the general solution of $\tan\theta = \sqrt{3}$. **[3]**

End of Worksheet • *Show all working clearly.*